CAPSTONE | BUILD & EVALUATE AN LLM-POWERED APPLICATION

Cross-Market Financial Quant & Filing Analyst Agent

A tool-augmented ReAct agent that computes exact ratios, cites real SEC filings, and abstains when it should

25.2% → 69.5%
Numerical accuracy

92.9%
Citation precision

0% → 80%
Trap abstention

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The Problem

Three bottlenecks make plain LLMs unsafe for filing analysis



Unreliable arithmetic

Next-token prediction can't reliably compute DuPont ROE, Altman Z-Score or FCF CAGR.



Temporal confusion

Standard RAG mixes 9-month cumulative, single-quarter and full-year guidance numbers.



Confident hallucination

Unknown companies or metrics get plausible falsehoods instead of an abstention.

Who It's For

Analysts who need exact numbers with verifiable provenance



Equity research & quant funds

Extract and validate quarterly metrics in seconds instead of hours of spreadsheet work.



Credit & risk officers

Altman Z-Score with SAFE / GREY / DISTRESS zones, linked to filing footnotes.

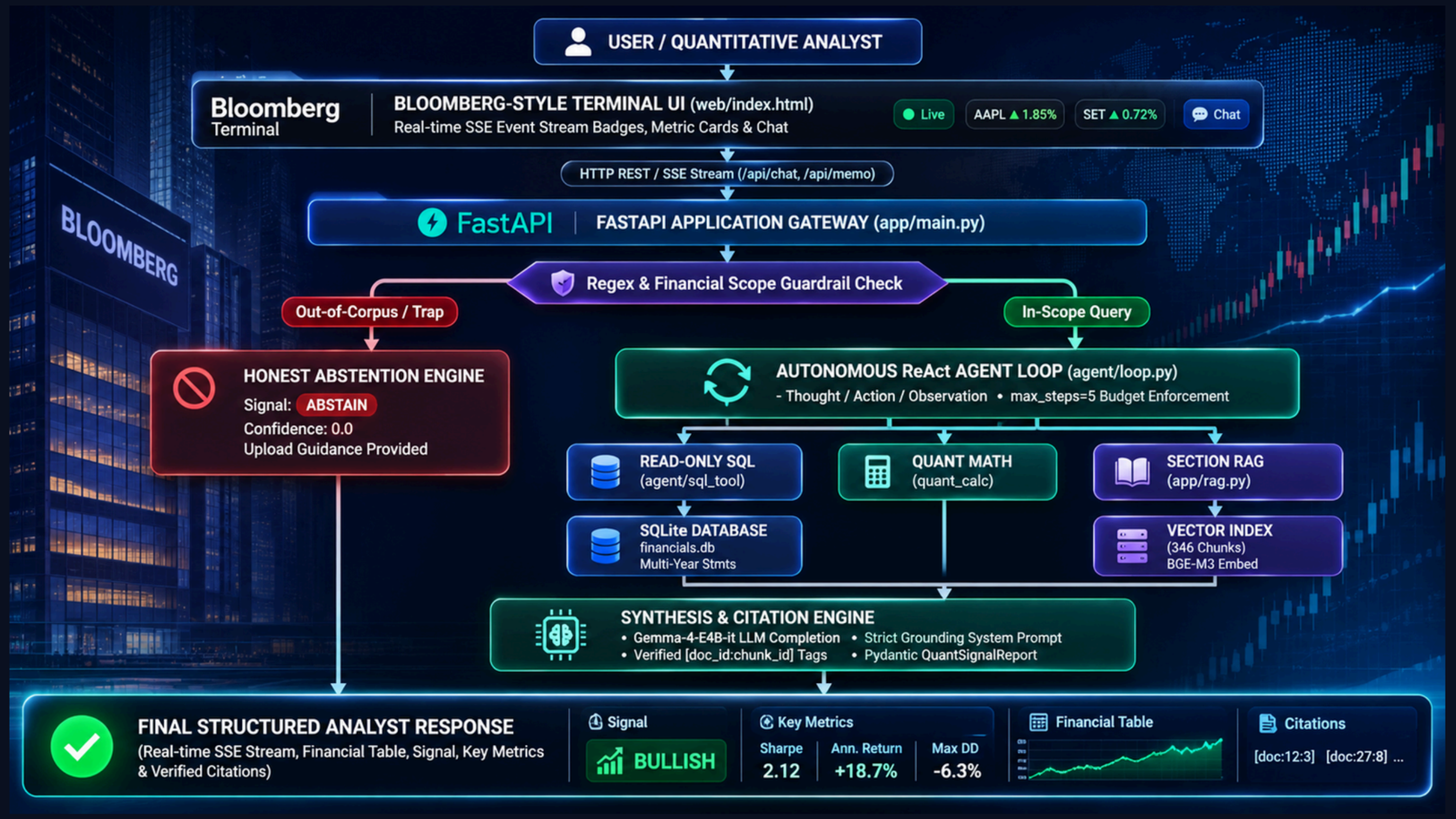


Corporate finance teams

Peer comparison with consistent accounting and paragraph-level [doc_id:chunk_id] citations.

System Architecture

A guardrail routes each query: abstain, or run a ReAct loop over three tools



1 Guardrail first

Regex and scope check sends out-of-corpus queries straight to ABSTAIN.

2 ReAct loop

Thought, action, observation with a max_steps=5 budget.

3 Cited output

Pydantic-validated report with verified citation tags.

Course Techniques Used

All 6 capability families from Lectures 1-15, each chosen for a reason

1 Prompting & structured output

Pydantic reports plus a repair parser keep LLM output machine-readable.

2 Backend & streaming

FastAPI + SSE show every tool call live, so each step is auditable.

3 Vector storage & RAG

Section-aware chunks, MIN_SCORE 0.30 and verified citation tags.

4 Rigorous evaluation

30 gold questions, 6 metrics, 2 baselines and an ablation grid.

5 Autonomous agent loop

ReAct with a 5-step budget; multi-ticker queries run per company.

6 Sandboxed SQL / code

Read-only SQLite and deterministic Python replace LLM arithmetic.

Evaluation Results: End-to-End Comparison

Tools lift accuracy, citations and abstention, but retrieval metrics fell

System	Recall@3	MRR@3	Faithful.	Num. acc.	Cite prec.	Trap abst.
B1 Zero-shot LLM	0.00%	0.00%	0.00%	25.22%	0.00%	0.00%
B2 Naive RAG	**28.00%**	**23.33%**	**78.00%**	47.41%	73.89%	0.00%
Proposed agent	10.00%	6.67%	53.33%	**69.45%**	**92.88%**	**80.00%**

What improved

Numerical accuracy +44 pts over zero-shot; citation precision 92.88%; trap abstention 0% to 80%.

What got worse

Recall@3 10% vs 28% for naive RAG; faithfulness 53% vs 78%. Retrieval is the weak link.

Failure Cases

Two failures found by the evaluation, both traced to a root cause

FAILURE 1 · Q18 & Q19

Period & run-rate mismatch

Query: *Compare BE and SPCX 2026 revenues and operating margins.*
What broke: Naive RAG pulled different quarters and mixed cumulative 9-month with single-quarter figures, confusing BE's Q2 $1,065.4M with its $4.4B-$4.6B full-year guidance.
Fix: Period-aligned statements in SQLite via query_financial_db.

FAILURE 2 · Q25

Over-aggressive thresholding

Query: *Across all 5 companies, state each DuPont ROE and guidance.*
What broke: A broad query diluted per-ticker similarity. Footnotes scored 0.32-0.48, so a high floor dropped 4 of 5 required sections.
Fix: MIN_SCORE=0.30 plus per-ticker tool runs in agent/loop.py.

Limitations & Next Steps

Retrieval quality is the most important thing still to fix

Limitations

- Recall@3 is only 10%; retrieval trails naive RAG
- Small test set: 30 questions, 5 companies, 10-Q only
- 1 of 5 trap queries was still answered (80% abstention)
- Filings must be clean Markdown; scanned tables unsupported
- No real-time market data; filings only

Next steps

- Rework retrieval: hybrid search and reranking, per-section thresholds
- Layout-aware vision models for image tables in PDFs
- Exchange data feeds for live price-to-earnings
- Bull vs bear sub-agent debate before the final memo
- Larger, multi-run evaluation with confidence intervals